Energy extension questions

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 A fully loaded fishing net has a mass of 500 kg. Calculate the work done raising it 100 m.

- $F = 500 \times 9.8 = 4900$ N
- $E_w = 4900 \times 100 = 490\,000$ J (490 kJ)

2 A weightlifter lifts equipment 1.2 m. The minimum work done is 600 J. Calculate the mass of the equipment.

- $F = E_w \div d = 600 \div 1.2 = 500$ N
- $m = 500 \div 9.8 = 51$ kg

3 A mechanic pulls a block-and-tackle rope 4 m with a force of 500 N. Calculate the work done.

- $E_w = 500 \times 4 = 2000$ J (2 kJ)

4 A 50 kg mass is raised 5 m by a rope wound round a pulley on a motor shaft. Calculate the work done by the motor.

- $F = 50 \times 9.8 = 490$ N
- $E_w = 490 \times 5 = 2450$ J (the pulley diameter isn't needed)

5 A 12 V car battery is rated at 35 amp hours (35 A for 1 hour). Calculate the energy it can supply.

- $E_e = 12 \times 35 \times 3600 = 1\,512\,000$ J (1.5 MJ)

6 A generator delivers 6 kJ of energy each second at 110 V. Calculate the current.

- $I = E \div (Vt) = 6000 \div (110 \times 1) = 55$ A

7 A 110 V, 30 A motor uses 1.98 MJ. For how long has it been running?

- $t = E \div (VI) = 1\,980\,000 \div (110 \times 30) = 600$ s (10 minutes)

8 A 230 V kettle takes 3.2 minutes to boil 1.8 litres of water from 17 °C. Calculate the current it draws, assuming no losses.

- $E_h = 4180 \times 1.8 \times 83 = 624\,492$ J
- $t = 192$ s
- $I = 624\,492 \div (230 \times 192) = 14$ A

9 50 kg metal ingots move along rollers at 0.5 m/s. Calculate the kinetic energy of each ingot.

- $E_k = 0.5 \times 50 \times 0.5^2 = 6.25$ J

10 A 50 kg cyclist has 2.5 kJ of kinetic energy. Calculate her speed, and the kinetic energy of her 30 kg bicycle.

- $v = \sqrt{(2 \times 2500 \div 50)} = 10$ m/s
- Bicycle: $0.5 \times 30 \times 10^2 = 1500$ J

11 Six people with a total mass of 900 kg ride an escalator at 0.5 m/s. Calculate the average kinetic energy of each person.

- Each person: $900 \div 6 = 150$ kg
- $E_k = 0.5 \times 150 \times 0.5^2 = 19$ J

12 A 10 kg radioactive-material container is crash-tested at 250 m/s in the USA and 50 m/s in the UK. Compare the kinetic energies.
- USA: $0.5 \times 10 \times 250^2 = 312\,500$ J
- UK: $0.5 \times 10 \times 50^2 = 12\,500$ J
- The US test has 25 times the kinetic energy (an increase of 2400%)

13 A conveyor lifts a 20 kg suitcase 3.5 m into an aircraft hold. Calculate the potential energy gained.
- $E_p = 20 \times 9.8 \times 3.5 = 686$ J

14 At what height does a 100 kg drum have 4 kJ of potential energy?
- $h = 4000 \div (100 \times 9.8) = 4.1$ m

15 A reservoir holds 1800 litres of water at a height of 260 m. Calculate the potential energy available.
- $E_p = 1800 \times 9.8 \times 260 = 4\,586\,400$ J = 4.6 MJ

16 A pile driver raised 5 m stores 9810 J. Calculate its weight and its mass.
- Weight = $E_p \div h = 9810 \div 5 = 1962$ N
- Mass = $1962 \div 9.8 = 200$ kg

17 A steeplejack has 2000 J of potential energy at 10 m. How much will he have at 15 m?
- E_p is proportional to height: $2000 \times 15 \div 10 = 3000$ J

18 A hall measures 8 m × 30 m × 10 m. Air has a density of 1.3 kg/m³ and c = 850 J/kg K. Calculate the heat needed to warm the air from 10 °C to 20 °C.
- Volume = 2400 m³, mass = $2400 \times 1.3 = 3120$ kg
- $E_h = 850 \times 3120 \times 10 = 26\,520\,000$ J = 27 MJ

19 A shower element gives 7 kJ of heat each second, warming water from 15 °C to 40 °C. Estimate the flow rate and state an assumption.
- $m = 7000 \div (4180 \times 25) = 0.067$ kg/s = 0.067 litres per second
- Assumes no heat is lost to the pipes or surroundings

20 54.3 kJ is supplied to 1.3 kg of oil (c = 2.72 kJ/kg K) at 1.5 °C. Calculate its final temperature.
- $\Delta T = 54\,300 \div (2720 \times 1.3) = 15.4$ °C
- Final temperature = 1.5 + 15.4 = 17 °C

21 6.3×10^5 J heats a statue (c = 3500 J/kg K) from 16 °C to 95 °C. Calculate its mass.
- $m = 630\,000 \div (3500 \times 79) = 2.3$ kg

22 2.4×10^4 J is added to 120 g of water at 19 °C. Calculate its final temperature.
- $\Delta T = 24\,000 \div (4180 \times 0.12) = 47.8$ °C
- Final temperature = 19 + 47.8 = 67 °C

23 How much heat is lost when 3 kg of water cools from 80 °C to 10 °C?
- $E_h = 4180 \times 3 \times 70 = 877\,800$ J = 880 kJ

24 How much heat is needed to raise 0.30 kg of aluminium from 30 °C to 150 °C?
- $E_h = 899 \times 0.3 \times 120 = 32\,364$ J = 32 kJ

25 Calculate the temperature change when 10.0 kg of water loses 232 kJ.
- $\Delta T = 232\,000 \div (4180 \times 10) = 5.6$ °C

26 Calculate the temperature change when 1.96 kJ is added to 500 g of copper.
- $\Delta T = 1960 \div (387 \times 0.5) = 10\ °C$

27 2.52×10^4 J is added to 2 kg of mercury, which reaches 130 °C. What was its starting temperature?
- $\Delta T = 25\,200 \div (138 \times 2) = 91.3\ °C$
- Starting temperature = $130 - 91.3 = 39\ °C$

28 A winch lifts a 500 kg pallet 10 m in 15 s. Calculate the minimum output power.
- $E_p = 500 \times 9.8 \times 10 = 49\,000$ J
- $P = 49\,000 \div 15 = 3300$ W (3.3 kW)

29 A 3 kW heater runs for one hour. Calculate the heat energy delivered.
- $E = P \times t = 3000 \times 3600 = 10\,800\,000$ J (10.8 MJ)

30 An 800 kg car reaches 70 km/h in one minute from rest. Calculate its kinetic energy and the average power needed.
- $v = 70 \div 3.6 = 19.4$ m/s
- $E_k = 0.5 \times 800 \times 19.4^2 = 151\,000$ J
- $P = 151\,000 \div 60 = 2500$ W

31 A 240 V, 10 A kettle takes 3 minutes to heat 0.5 kg of water from 20 °C to 100 °C. Find the energy supplied, the heat used and the efficiency.
- $E_{in} = 240 \times 10 \times 180 = 432\,000$ J
- $E_h = 4180 \times 0.5 \times 80 = 167\,200$ J
- $\eta = 167\,200 \div 432\,000 = 39\%$

32 Water falls 725 m through a hydroelectric station at 5125 kg/s. The station is 72% efficient. Calculate the electrical energy produced each second.
- $E_p = 5125 \times 9.8 \times 725 = 36\,400\,000$ J
- $E_{out} = 0.72 \times 36.4$ MJ = 26 MJ

33 A lift raises 200 kg through 10 m in 2 minutes. The motor uses 12 MJ per hour. Calculate the efficiency of the lift.
- Energy in over 2 minutes = 12 MJ ÷ 30 = 400 000 J
- $E_p = 200 \times 9.8 \times 10 = 19\,600$ J
- $\eta = 19\,600 \div 400\,000 = 4.9\%$

34a) Calculate the current drawn.
- $I = V \div R = 240 \div 12 = 20$ A

34b) Calculate the heat energy transferred to the water.
- $E_h = 4180 \times 0.15 \times 60 = 37\,620$ J

34c) Calculate the time taken if the system is 100% efficient.
- $P = VI = 4800$ W; $t = 37\,620 \div 4800 = 7.8$ s

34d) It actually takes 10 s. Calculate the percentage of energy lost.
- $E_{in} = 4800 \times 10 = 48\,000$ J
- Lost = $48\,000 - 37\,620 = 10\,380$ J = 22%